CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 2: Compound Interest (Without Using Formula)

Topic-wise short notes with solved examples

Finding compound interest year by year using the simple interest formula, with interest compounded yearly or half-yearly, and comparing compound interest with simple interest.

1 Simple interest

- **Principal** P: the money borrowed or lent. **Rate** $R\%$ per annum. **Time** T years.

$$\text{S.I.} = \frac{P \times R \times T}{100}, \qquad \text{Amount} = P + \text{S.I.}$$

- With simple interest the principal stays the same for the whole period, so the interest is the same every year.

Example 1.1 Find the simple interest on ₹8000 at 6% per annum for 3 years.

SOLUTION

$$\frac{8000 \times 6 \times 3}{100}$$

₹1440

2 Compound interest

- With **compound interest** the interest of each period is added to the principal. The **amount at the end of one year becomes the principal for the next year**.
- Compound interest (C.I.) = final amount − original principal.
- **Method**: find the interest for the first year by the S.I. formula with $T = 1$; add it to get the amount; use that amount as the new principal; repeat.

Year	Principal (₹)	Interest at 10% (₹)	Amount (₹)
1	5000	500	5500
2	5500	550	6050
3	6050	605	6655

Example 2.1 Find the amount and the compound interest on ₹5000 for 3 years at 10% per annum, compounded yearly.

SOLUTION

From the table, the amount after 3 years is ₹6655.

Amount ₹6655; C.I. $= 6655 - 5000$, that is ₹1655

Example 2.2 Find the compound interest on ₹12,000 for 2 years when the rate is 8% for the first year and 10% for the second.

SOLUTION

First year: interest $= 960$, amount $= 12960$.

Second year: interest $= 1296$, amount $= 14256$.

C.I. ₹2256

3 Useful facts

- For the **first year**, C.I. and S.I. are **equal**.
- The interests of successive years grow: interest of any year $=$ interest of the previous year $+$ interest on that interest.
- The **amounts** of successive years are in a fixed ratio: each is $\left(1 + \frac{R}{100}\right)$ times the one before.
- **Difference of the C.I. of two successive years** $=$ one year's interest on the earlier year's C.I. So

$$R = \frac{\text{difference between the C.I. of two successive years}}{\text{C.I. of the earlier year}} \times 100$$

- **Difference of the amounts of two successive years** is the interest on the earlier amount. So $R = \frac{\text{difference of the amounts}}{\text{earlier amount}} \times 100$.

Example 3.1 The compound interest on a sum is ₹1200 in the first year and ₹1320 in the second year. Find the rate and the sum.

SOLUTION

Difference $= 120$, which is one year's interest on ₹1200. So $R = \frac{120}{1200} \times 100 = 10\%$.

First-year interest: $\frac{P \times 10}{100} = 1200$.

Rate 10%; sum ₹12,000

Example 3.2 A sum amounts to ₹9680 in 2 years and to ₹10,648 in 3 years at compound interest. Find the rate and the sum.

SOLUTION

Interest for the third year $= 10648 - 9680 = 968$ on ₹9680, so $R = \dfrac{968}{9680} \times 100 = 10\%$.

Amount after 1 year $= 9680 \div 1.1 = 8800$; sum $= 8800 \div 1.1$.

Rate 10%; sum ₹8000

4 Comparing C.I. and S.I.

- For more than one year at the same rate, **C.I. is greater than S.I.**
- For 2 years, C.I. $-$ S.I. $=$ one year's interest on the first year's interest.

Example 4.1 Find the difference between the C.I. and the S.I. on ₹6000 for 2 years at 5% per annum.

SOLUTION

S.I. $= \dfrac{6000 \times 5 \times 2}{100} = 600$.

C.I.: first year 300; second year 5% of 6300 $= 315$; total 615.

₹15

Example 4.2 The difference between the C.I. and the S.I. on a sum for 2 years at 8% is ₹64. Find the sum.

SOLUTION

The difference is 8% of the first year's interest: $0.08 \times (0.08P) = 64$.

$0.0064P = 64$

₹10,000

5 Interest compounded half-yearly

- When interest is compounded **half-yearly**, the rate for each half-year is **half** the annual rate, and the number of periods is **twice** the number of years.
- Each half-year's interest $= \dfrac{P \times R \times 1}{100 \times 2}$, where P is the principal for that half-year.

Example 5.1 Find the compound interest on ₹10,000 for 1 year at 8% per annum, compounded half-yearly.

SOLUTION

Rate per half-year $= 4\%$. First half-year: interest 400, amount 10400.

Second half-year: interest 4% of 10400 $= 416$, amount 10816.

₹816

Example 5.2 Find the amount on ₹8000 for $1\frac{1}{2}$ years at 10% per annum, compounded half-yearly.

SOLUTION

Three half-years at 5%: $8000 \to 8400 \to 8820 \to 9261$

₹9261

6 Repayments and growth

- If part of a loan is **repaid** at the end of a year, subtract the repayment from the amount. The balance is the principal for the next year.
- The same year-by-year method applies to anything that grows by a fixed percentage: population, value of property, production.
- For a **fall** in value (depreciation), subtract the year's decrease instead of adding.

Example 6.1 A man borrows ₹10,000 at 10% per annum compound interest. He repays ₹4000 at the end of each of the first two years. How much does he owe at the end of the third year?

SOLUTION

Year 1: $10000 + 1000 - 4000 = 7000$.

Year 2: $7000 + 700 - 4000 = 3700$.

Year 3: $3700 + 370$

₹4070

Example 6.2 A machine worth ₹20,000 loses 10% of its value every year. Find its value after 2 years.

SOLUTION

Year 1: $20000 - 2000 = 18000$. Year 2: $18000 - 1800$.

₹16,200

7 Common mistakes

- Using the original principal for every year. Each year's principal is the previous year's amount.
- Giving the amount when the compound interest is asked. C.I. = amount − original principal.
- For half-yearly compounding, halving the rate but forgetting to double the number of periods.
- Taking the rate from the difference of two successive years' interests divided by the **later** year's interest.
- Subtracting a repayment before adding that year's interest.

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